EMPIRICAL EVALUATION · LLM CODE GENERATION · C# / .NET

Benchmarking the Titans



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A Multi-Dimensional Evaluation of LLM Code Generation Quality in the .NET Ecosystem

GPT · Gemini · Claude · Grok

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MOTIVATION

Two blind spots in how we evaluate LLM code

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Pass@k hides quality

A function with cyclomatic complexity 19 and one with complexity 2 , both passing every test , are identical under Pass@k, yet very different engineering artifacts.

Python monoculture

HumanEval, MBPP and peers focus almost entirely on Python. Enterprise-critical C#/.NET has no unified correctness + quality + runtime benchmark.

The real question: not just *“does it pass?”* , but is it maintainable, efficient, and stylistically sound?

OUR CONTRIBUTION

A unified, 3-dimensional C# evaluation framework

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Correctness

Automated unit testing via .NET Reflection

Static Quality

Roslyn AST: complexity, nesting, naming, style

Runtime Efficiency

Adversarial BenchmarkDotNet profiling

Research Questions

RQ1 How does functional correctness compare across the four models?

RQ2 Does task difficulty , not the model , drive structural complexity?

RQ3 Are correctness and composite quality statistically related, or orthogonal?

RQ4 How do solutions compare in runtime time & memory under adversarial profiling?

THE FRAMEWORK

A four-phase automated pipeline

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PHASE 1



Dataset Prep

85 HumanEval tasks adapted to C#; prompts + benchmark harnesses prepared

PHASE 2



Code Generation

4 LLMs queried in parallel @ temp = 0 → 340 solutions

PHASE 3



Dynamic Eval

3 passes: correctness · Roslyn AST · BenchmarkDotNet

PHASE 4



Result Collection

340 rows consolidated; composite score + RQ3 analysis

85 C# tasks

4 LLMs

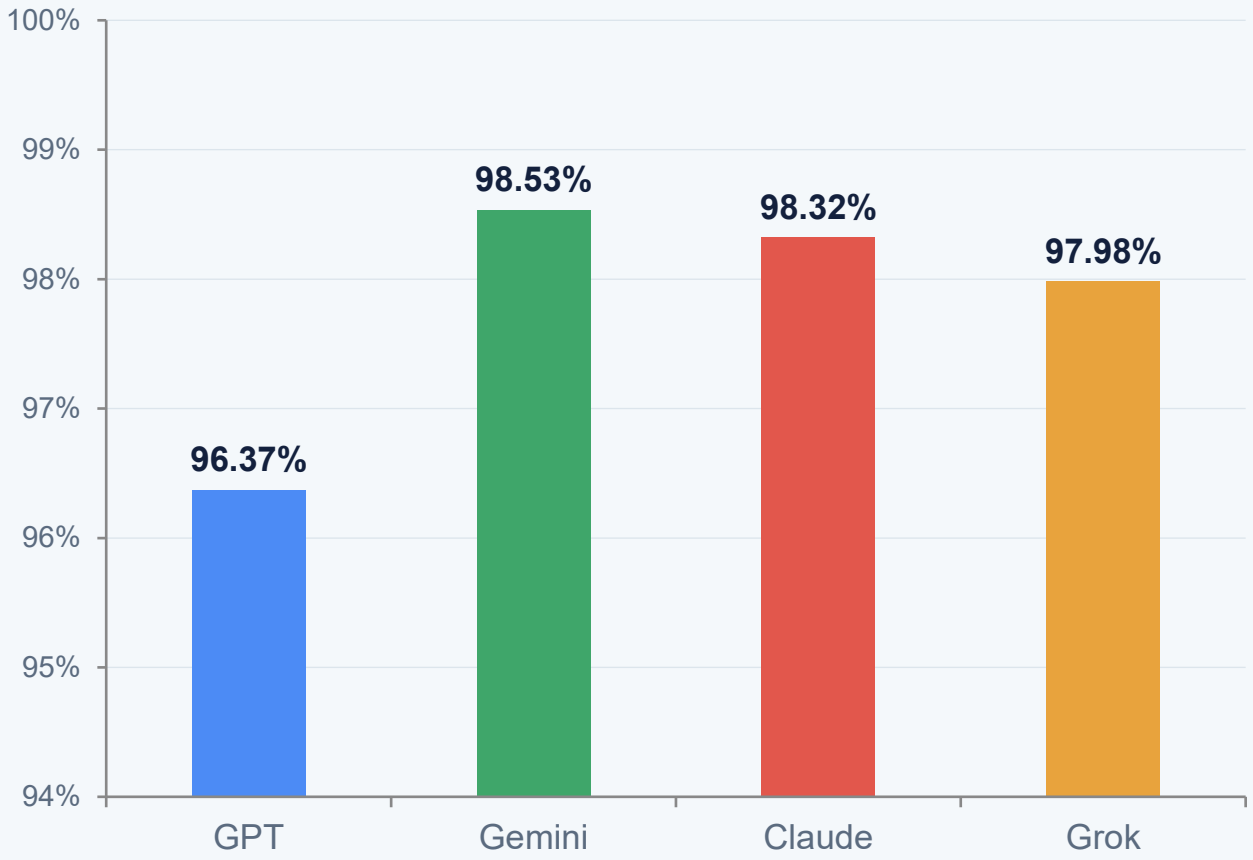
340 solutions

3 quality dimensions

HumanEval → C# adaptation · class Solver structure enforced · DeepSeek-V3 generates worst-case benchmark inputs (excluded from the evaluated pool to avoid bias)

All four cluster high, but GPT is bimodal

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Gemini leads

98.53% mean · only 2 failures — the most reliable on this benchmark.

GPT is volatile

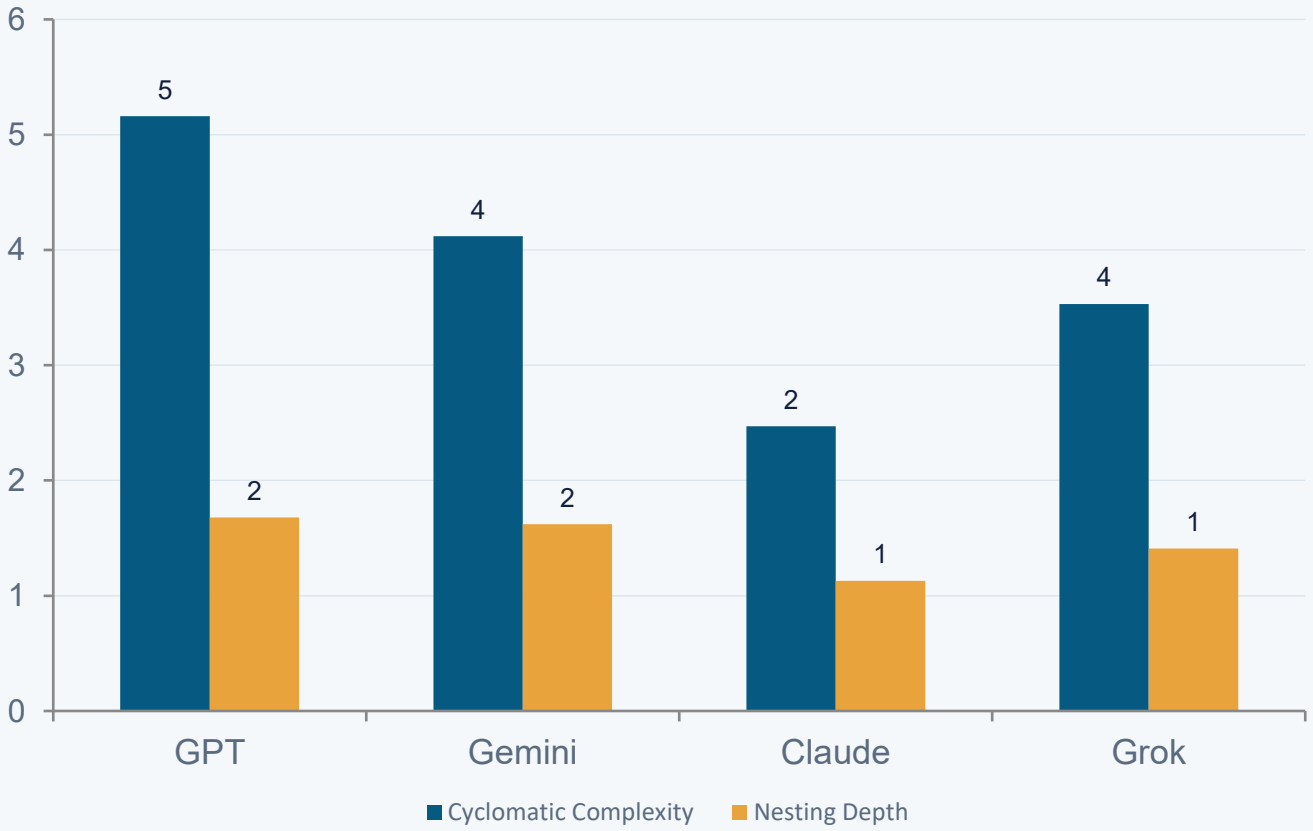
Lowest mean (96.37%) and highest variance (SD 15.37 vs Claude 9.46): fully-correct or substantially-failed.

Shared limits

csharp_57 & csharp_64 trip all models identically — hints of shared training data.

Difficulty — not the model — drives complexity

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r = 0.63–0.69

cross-model complexity agreement on identical tasks

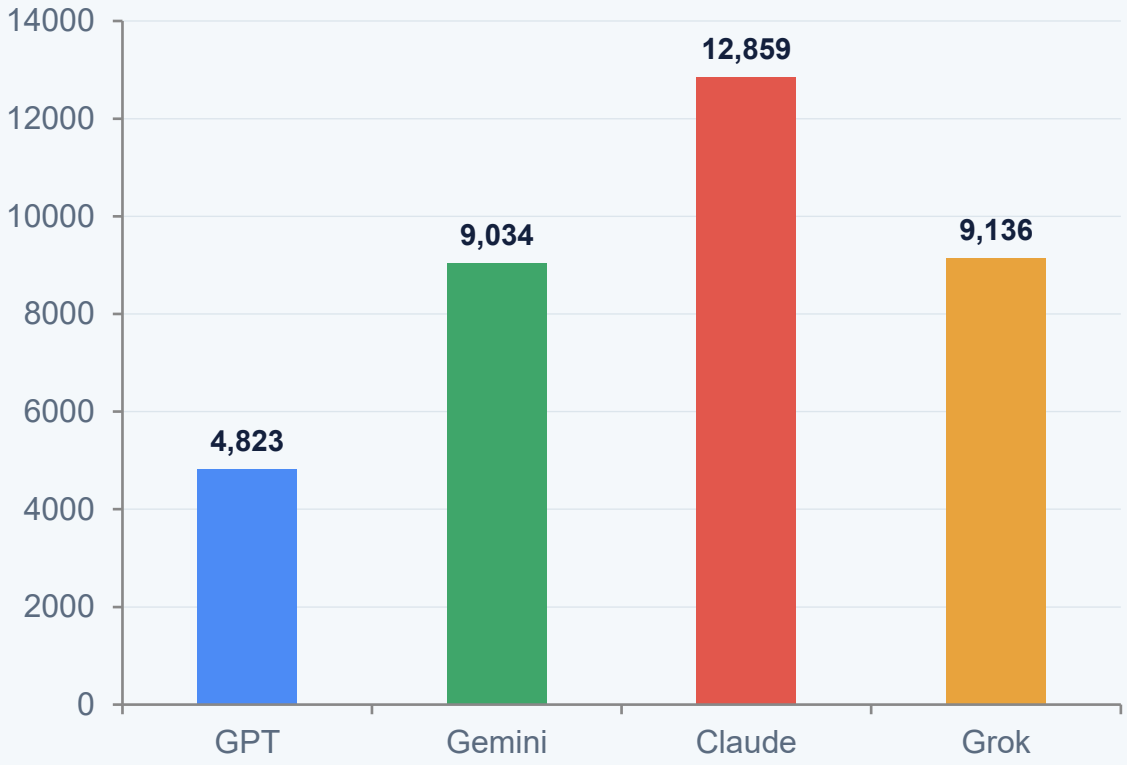
Claude writes the simplest code

Complexity 2.47 — roughly half of GPT's 5.16. But the agreement across models shows recursion / multi-branch tasks force complexity on everyone. Compare complexity only within the same task.

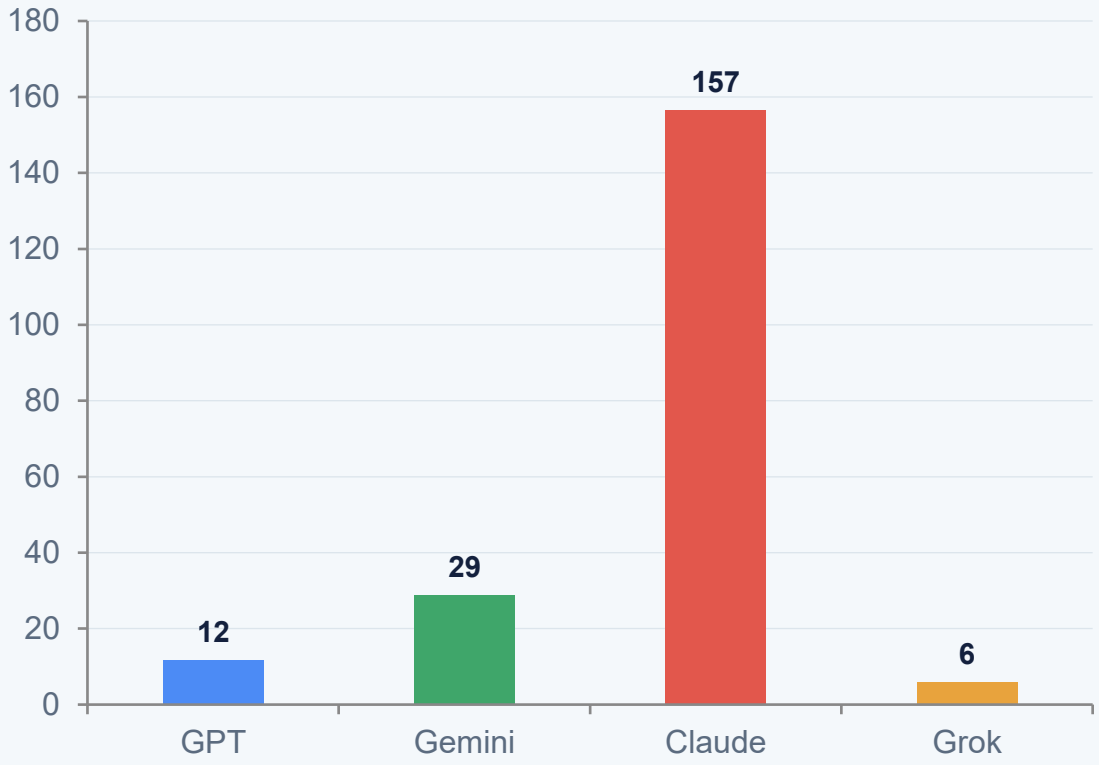
Speed and memory diverge sharply

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Median Execution Time (ns)



Median Memory Allocation (MB)



GPT fastest (4,823 ns). Claude's outliers — csharp_12 (~2,500× slower) and csharp_16 (~3.94 GB) — reveal a pathological, data-intensive strategy on two tasks.

RQ3 · THE CENTRAL FINDING

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Correctness and quality are orthogonal

$$r = 0.075$$

Pearson correlation between functional correctness and composite quality (n = 340)

Ranking reversal

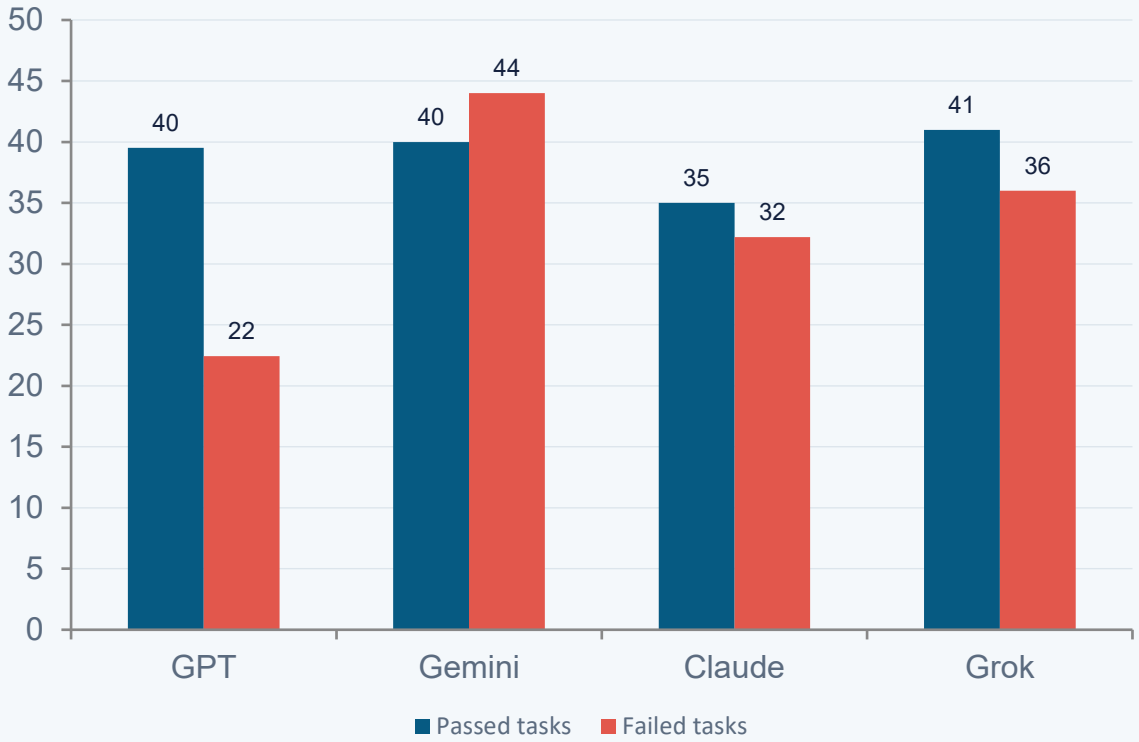
By Correctness			By Quality Score		
1.	Gemini	98.53	1.	Grok	40.80
2.	Claude	98.32	2.	Gemini	40.05
3.	Grok	97.98	3.	GPT	38.31
4.	GPT	96.37	4.	Claude	37.69

Pass@k rankings are statistically uninformative about code quality. Correctness and structural quality must be reported as independent, co-equal dimensions.

No model wins everywhere — pick by priority

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GPT's quality collapses on failed tasks



Gemini — Most reliable — highest correctness

Grok — Most balanced — best composite quality, leanest memory

Claude — Simplest code, but highest memory footprint (156.6 MB)

GPT — Fastest runtime, but bimodal quality collapse on failure

What we control for — and what we don't

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Construct

r = 0.075 is partly self-referential (Norm_C is one of 5 score components). Robustness check excluding correctness is future work.



Internal

temp = 0 mitigates but doesn't eliminate API non-determinism. Median absorbs Claude's runtime outliers.



External

85 single-file HumanEval tasks aren't enterprise .NET. Orthogonality may not generalize to other languages.



Conclusion

85 tasks limit power — mitigated by analyzing all 340 records and reporting SDs throughout.

KEY TAKEAWAYS

What to remember

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Correctness ≠ quality

r = 0.075 across 340 solutions: Pass@k tells you almost nothing about maintainability or efficiency.

Difficulty drives complexity

Cross-model agreement (r = 0.63–0.69) — compare complexity only within the same task.

Quality has many axes

Simple code can be memory-hungry (Claude); fast code can be structurally messy (GPT). No model wins all.

C#/.NET needs its own bench

A unified, automated, multi-dimensional pipeline — fully reproducible and dataset-agnostic.

Thank you

Correctness and quality are orthogonal — evaluate LLM code on both.

github.com/mohammadmehdighalandarian/llm4se

Harness · adapted task specs · raw results — fully reproducible



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